

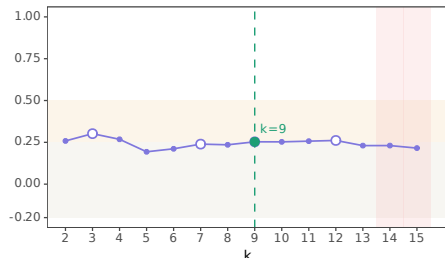
k (number of cluster) optimization (93 genera x 11 age groups)

k = 9 is the chosen number of clusters; teal disc + dashed teal vline mark k=9 on each panel.

A

Silhouette: local maxima

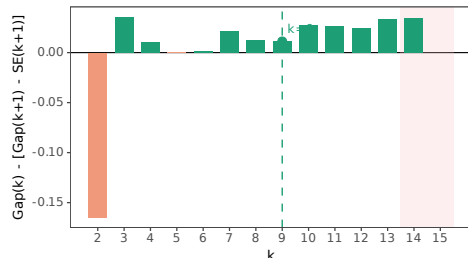
$s_bar(k)$ in $[-1, 1]$. Open circles = local maxima at $k = 3, 7, 9, 12$. Background color: Green = local maxima, Red = local minima, Blue = local maxima, Yellow = local minima, Purple = local maxima, Orange = local minima, Pink = local maxima, Gray = local minima.



B

Gap statistic (Tibshirani SE-rule)

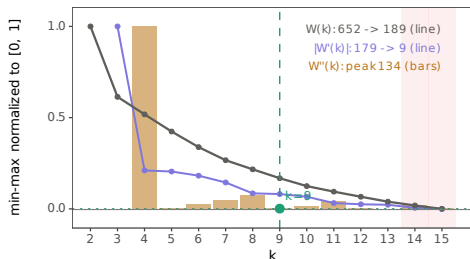
Gap(k) - [Gap(k+1)]



C

W (within-cluster dispersion) and its derivatives Resolution ceiling (first-touch-the-floor)

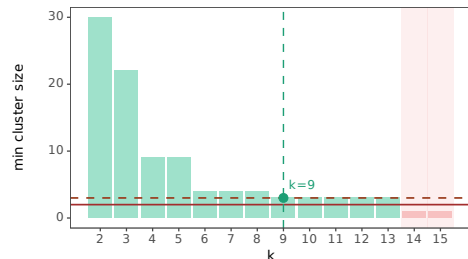
W(k) = within-cluster dispersion (same quantity the gap statistic compares) Resolution ceiling (first-touch-the-floor) is the first k such that W(k) is less than or equal to W(k+1). The resolution ceiling is the first k such that W(k) is less than or equal to W(k+1).



D

Resolution ceiling (first-touch-the-floor)

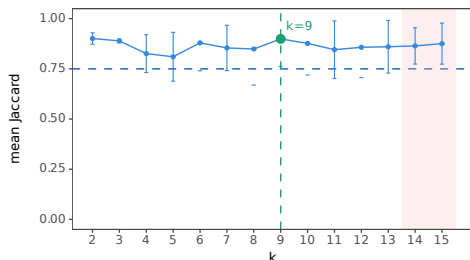
Resolution ceiling (first-touch-the-floor) is the first k such that W(k) is less than or equal to W(k+1). The resolution ceiling is the first k such that W(k) is less than or equal to W(k+1).



E

Stability (bootstrap consensus)

Bootstrap mean jaccard +/- 1 sd vs k. Dashed line: stable >= 0.75. At k=9 stability holds (mean jaccard = 0.899) and persists for k=10, 11, 12, 13, 14, 15.



Gray-washed columns = infeasible (min_size < 2). egg::theme_artide(base_size = 5).